

Shreya Patha

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Profile

AI Engineer with hands-on experience building Python backend applications, integrating LLM APIs, and developing multi-agent workflows and autonomous AI systems. Experienced with FastAPI, REST APIs, prompt engineering, structured outputs, and agentic pipelines using tools such as Groq, Claude, LangGraph, and PostgreSQL.

Education

Bachelor of Technology in Computer Science Engineering in Artificial intelligence and machine learning 11/2022 – 05/2026
TKR College of Engineering and Technology

Certificates

Workshop on Generative AI and Automation Tools (Outskill) | Business Analytics(ThinkMates)
Prompt Engineering Fundamentals Deversigns 1.0 Hackathon

Languages

– Telugu Hindi | English

Projects

Interview — Autonomous Multimodal Technical Assessment Platform,

Python, FastAPI, Next.js 16, TypeScript, Gemini 2.0 Flash, ReportLab, AES-256 Fernet, Pytest

- Architected a multimodal interview platform parsing resumes/IDs to dynamically generate 3-stage adaptive curricula via Gemini 2.0 Flash.
- Engineered privacy-first telemetry measuring real-time poise and speech cadence; secured records using AES-256 Fernet encryption and automated PII scrubbing.
- Built server-side ReportLab PDF dossier generation and achieved 100% subsystem test pass rate across 89 Pytest suites.

EduBridge - AI Human-Like School Assistant, Python, FastAPI, NLU, RBAC, Pydantic, STT/TTS, Multilingual AI

- Developed a role-based conversational AI assistant supporting chat, voice interaction, and file-synced avatars across 11 Indian languages for students, teachers, and administration.
- Implemented NLU intent/entity extraction, strict Pydantic validation, deterministic tool routing, and role-spoofing guards backed by 109 automated unit and integration tests.

AI Research Assistant — Multi-Agent Research Pipeline, FastAPI, React, LangGraph, Groq, Tavily, PostgreSQL, WebSockets

- Built a multi-agent research assistant orchestrating search, summarization, and fact-checking agents via LangGraph, integrating Groq and Tavily APIs into REST and WebSocket endpoints with live progress streaming.
- Designed a PostgreSQL-backed persistence layer with Pydantic-validated structured outputs for research history and report generation.

Experience

AI/ML Team Lead — UpToSkills

02/2026 – 05/2026

- Led a cross-functional cohort of 8–15 AI/ML, Python, and Data Analytics trainees as Subject Captain, improving on-time task submission by ~30%, by restructuring attendance tracking, task coordination, and cross-team communication workflows.
- Automated cohort performance reporting, cutting manual review time by ~5 hours/week, by building a Python backend pipeline that ingests CSV performance data, calls the Claude API to generate structured JSON coaching reports, and flags low-score weekly trends.
- Reduced repetitive support queries handled by mentors by ~35%, by developing SkillNova, an automated technical support assistant using the Groq API and LLM-driven query resolution.
- Kept the coaching-report pipeline dependency-free and easier to audit, with zero external SDK dependencies, by integrating the Claude API directly via Python's urllib — handling authentication headers, request formatting, and response parsing with raw HTTP requests.

Skills

LLM/AI Integration: Claude API, OpenAI API, Groq API, Prompt Engineering

Backend & APIs: REST APIs, JSON, FastAPI, WebSockets, Pydantic

Databases: PostgreSQL, MySQL, ChromaDB

Data & ML: Pandas, NumPy, Scikit-learn, TensorFlow, NLP, Deep Learning, EDA

Visualizations: Power BI, Matplotlib, Seaborn, Excel